

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z} dz$$

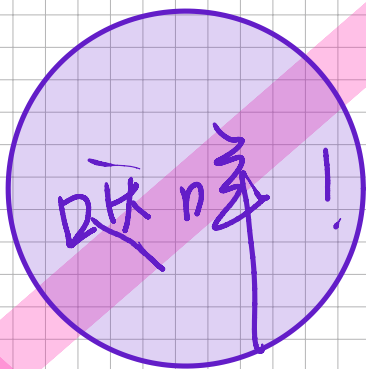
ccw once around 0

$$\text{Say } f(z) = C_0 + C_1 z + C_2 z^2 + \dots$$

$$\Rightarrow \frac{f(z)}{z} = \frac{C_0}{z} + C_1 + C_2 z + \dots$$

$$\Rightarrow \oint_C \frac{f(z)}{z} dz = \oint_C \frac{C_0}{z} + 0 + 0 + \dots$$

$$= 2\pi i C_0 = 2\pi i f(0)$$



FACI f holomorphic

$$\cdot \frac{d^n}{dz^n} f(w) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-w)^{n+1}} n! dz$$

f 全纯, 复可导 $\Rightarrow$ 无限可导

$\Rightarrow f$ differentiable (once), i.e., $f \in C^1 \Rightarrow f \in C^\infty$

• Even better, f has Taylor Series. $f(w) = \sum_{n \geq 0} \frac{f^{(n)}(z_0)}{n!} w^n$ (within ROC)

Leibniz Integral Rule

$$\tilde{f}(x) = \int_{a(x)}^{b(x)} f(x, t) dt$$

We want $\frac{d\tilde{f}}{dx}$

How?

Define a support function $G(x, u, v) = \int_u^v f(x, t) dt$.

Then, $\tilde{f}(x)$ can be represented as: $\tilde{f}(x) = G(x, a(x), b(x))$, where $u = a(x)$, $v = b(x)$

Then, we take the derivative of $\tilde{f}(x)$.

By Chain Rule, we have

$$\frac{d\mathcal{E}}{dx} = \frac{d\mathcal{E}(x, u, v)}{dx} = \frac{\partial \mathcal{E}}{\partial x} \frac{du}{dx} + \frac{\partial \mathcal{E}}{\partial u} \frac{du}{dx} + \frac{\partial \mathcal{E}}{\partial v} \frac{dv}{dx}$$

Since $u = a(x)$, $v = b(x)$, we have $\frac{du}{dx} = a'(x)$, $\frac{dv}{dx} = b'(x)$

$$\frac{\partial \mathcal{E}}{\partial v} = \frac{\partial}{\partial v} \int_u^v f(x, t) dt = f(x, v) \quad (\text{Fundamental Theorem of Calculus}).$$

$$\text{Similarly, } \frac{\partial \mathcal{E}}{\partial u} = \frac{\partial}{\partial u} \int_u^v f(x, t) dt = -\frac{\partial}{\partial u} \int_u^u f(x, t) dt$$

refer to as const $= -f(x, u) dt$

$$\frac{\partial \mathcal{E}}{\partial x} = \frac{\partial}{\partial x} \int_u^v f(x, t) dt = \int_u^v \frac{\partial}{\partial x} f(x, t) dt$$

$$\begin{aligned} \text{Thm. } \frac{d\mathcal{E}}{dx} &= \frac{\partial \mathcal{E}}{\partial x} \frac{du}{dx} + \frac{\partial \mathcal{E}}{\partial u} \frac{du}{dx} + \frac{\partial \mathcal{E}}{\partial v} \frac{dv}{dx} \\ &= \int_u^v \frac{\partial}{\partial x} f(x, t) dt \frac{dx}{dx} + f(x, v) \frac{dv}{dx} - f(x, u) \frac{du}{dx} \\ &= \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x, t) dt + f(x, b(x)) b'(x) - f(x, a(x)) a'(x) \end{aligned}$$

$$\text{Thus, } \frac{d}{dx} \int_{a(x)}^{b(x)} f(x, t) dt = f(x, b(x)) b'(x) - f(x, a(x)) a'(x) + \int_{a(x)}^{b(x)} \frac{\partial f}{\partial x} dt$$

In the case where $a(x)$ and $b(x)$ are constant, this formula reduces to the simpler form

$$\frac{d}{dx} \int_a^b f(x, t) dt = \int_a^b \frac{\partial}{\partial x} f(x, t) dt.$$

→ This simpler statement is known as **Leibniz Integral Rule**.

$$\text{Pf: } \frac{d^n}{dz^n} f(w) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-w)^{n+1}} n! dz$$